

Rafael Ferreira de Menezes

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Research Summary

Computational chemist and materials scientist working on electrochemical interfaces and energy-storage materials. Combines quantum chemistry, classical and reactive molecular dynamics, and machine-learned interatomic potentials with operando synchrotron X-ray diffraction, reflectivity, and spectroscopy at the Advanced Photon Source and NSLS-II. Current interests extend this toolkit to planetary and extraterrestrial materials, ices, and power systems for space missions. Ph.D. expected Spring 2027.

Education

University of Colorado Boulder

Ph.D. in Chemical Engineering (expected May 2027)
– Advisors: [Kayla G. Sprenger](#) and [Michael F. Toney](#)

Boulder, CO, USA

Aug 2022 – Present

University of Colorado Boulder

M.S. in Chemical Engineering

Boulder, CO, USA

Aug 2022 – May 2024

University of Brasilia

B.S. in Physics
– Advisor: [Ricardo Gargano](#)

Brasilia, DF, Brazil

Mar 2017 – Nov 2021

Research Experience

University of Colorado Boulder

Graduate Research Assistant, Sprenger and Toney groups

Boulder, CO, USA

Aug 2022 – Present

- Model cathodes, electrolytes, and their interfaces in lithium-ion, lithium-metal, sodium-ion, and lithium-sulfur batteries using DFT and coupled-cluster calculations, classical and reactive (ReaxFF) molecular dynamics, and hybrid molecular dynamics/Monte Carlo sampling.
- Design and run operando synchrotron experiments (X-ray diffraction, reflectivity, and absorption spectroscopy) at the Advanced Photon Source and NSLS-II to follow crystal-structure evolution, phase transitions, and oxidation states during cycling.
- Contribute to two DOE Vehicle Technologies Office projects: mechanistic modeling of Li-S batteries and mechanistic studies of the cathode-electrolyte interface.
- Run large-scale simulations on the HPC systems of the National Laboratory of the Rockies (NLR, formerly NREL).

Lawrence Livermore National Laboratory

Academic Collaboration Program Intern

Livermore, CA, USA

Jun – Aug 2024

- Developed machine-learned interatomic potentials for lithium-ion battery electrolytes.

Synchrotron and HPC User Facilities

Experimental user: APS (Argonne), NSLS-II (Brookhaven); HPC user: NLR

USA

2023 – Present

- High-flux X-ray reflectivity of layered structures at battery interfaces (APS) and operando X-ray diffraction of cathode reconstruction during cycling (NSLS-II).

University of Brasilia

Undergraduate Researcher, Gargano group

Brasilia, DF, Brazil

Mar 2020 – Jun 2022

- Studied electronic and dynamic properties of functionalized graphene quantum dots and their adsorption of pollutant gases (CO, NO₂, SO₂, NH₃).
- Characterized van der Waals interactions in systems containing superheavy elements (oganesson).

Brazilian National Council for Scientific and Technological Development (CNPq)

Scientific Initiation Fellow

Brasilia, DF, Brazil

Mar 2019 – Nov 2021

- Computed the dynamics, spectroscopy, and lifetimes of weakly bound complexes with ab initio quantum chemistry, in collaboration with the Università degli Studi di Perugia (Italy).

Scientific and Technological Enterprises Foundation (FINATEC)

Intern

Brasilia, DF, Brazil

Jul – Dec 2018

- Built a machine-learning program to organize and classify documents for the Brazilian Supreme Court (STF).

Skills

Methods	DFT and coupled-cluster quantum chemistry, TDDFT; classical, reactive (ReaxFF), and machine-learned-potential molecular dynamics; hybrid molecular dynamics/Monte Carlo; QTAIM and NBO bonding analysis; simulated annealing of amorphous carbons
Software	LAMMPS, GROMACS, CP2K, Quantum ESPRESSO, Gaussian, ORCA, SIESTA, Psi4, Multiwfn, AMS; Python scientific stack (NumPy, SciPy, matplotlib, scikit-learn); Linux, Slurm, Git
Experimental	Operando and in situ synchrotron X-ray diffraction, reflectivity, and absorption spectroscopy; coin-cell assembly and electrochemical testing; X-ray data reduction
Languages	Portuguese (native), English (fluent)
Certificates	Google Cybersecurity Professional Certificate (2024)

Publications

Google Scholar: 119 citations, h-index 6 (September 2026). Own name in bold.

Journal Articles

- [1] Lietzke, E. E., Saeb, D., Aldrich, E. C., **de Menezes, R. F.**, Bruce, K. D., Sprenger, K. G. (2026). Uncovering the biological mechanisms of TREM2 with molecular simulations: a comprehensive review and perspective. *Progress in Biophysics and Molecular Biology* 201, 189–211. doi:10.1016/j.pbiomolbio.2026.06.002
- [2] Pimenta, B. G. A., Vasconcelos, R. C., de Oliveira Neto, P. H., **de Menezes, R. F.**, Sprenger, K. G., Gargano, R. (2025). Stability and induced magnetism by edge modification of HfS₂ nanoribbons. *Langmuir* 41(31), 21066–21072. doi:10.1021/acs.langmuir.5c02830

- [3] de Oliveira, R. M., de Sousa, L. N., Gregorio, A. V., **de Menezes, R. F.**, Sprenger, K. G., de Macedo, L. G. M., Pirani, F., Gargano, R. (2025). The interatomic bond between Cd and noble gas atoms: the nature of the intermolecular forces, the stability of the adducts, their spectroscopic and thermodynamical properties. *Computational and Theoretical Chemistry* 1249, 115238. doi:10.1016/j.comptc.2025.115238
- [4] Kitsu Iglesias, L., Marks, S. D., Rampal, N., Antonio, E. N., **Ferreira de Menezes, R.**, Zhang, L., Olds, D., Weitzner, S. E., Sprenger, K. G., Wan, L. F., Toney, M. F. (2025). Microstructure-dependent sodium storage mechanisms in hard carbon anodes. *Small* 21(30), 2505561. doi:10.1002/sml.202505561
- [5] de Araujo Oliveira, A. L., Medeiros, A. K., Oliveira, R. M., **Ferreira de Menezes, R.**, Sprenger, K. G., Pirani, F., Gargano, R. (2025). Ammonia-noble gas systems: insight on the nature of the intermolecular interaction, spectroscopic and thermodynamical properties. *Advances in Quantum Chemistry* 92, 121–139. doi:10.1016/bs.aiq.2025.06.004
- [6] Ou, W., Marks, S. D., **de Menezes, R. F.**, He, R., Zhang, Z., Sindt, C., Thurston, J., Jaye, C., Cowie, B., Thomsen, L., Zhuo, Z., Guo, J., Yang, W., Dong, Z., Tenent, R., Sprenger, K. G., Toney, M. F. (2025). Unveiling the mechanism of Mn dissolution through a dynamic cathode-electrolyte interphase on LiMn₂O₄. *Advanced Energy Materials* 15(22), 2404652. doi:10.1002/aenm.202404652
- [7] He, R., McDonough, L., Seitz, L., Ou, W., Marks, S. D., **Ferreira de Menezes, R.**, Allan-Cole, E., Luo, H., Toney, M. F., Sprenger, K. G., Zhou, M., Tenent, R. C. (2025). In situ characterization of the oxidation behavior of carbonate-based electrolytes for lithium-ion batteries by scanning electrochemical microscopy. *ACS Electrochemistry* 1(4), 494–503. doi:10.1021/acselectrochem.4c00106
- [8] de Jesus, G. C. C., Vasconcelos, R. C., do Vale, L. B., **de Menezes, R. F.**, Sprenger, K. G., Gargano, R. (2025). Electronic and magnetic properties of manganese bromide monolayers. *Langmuir* 41(9), 5877–5883. doi:10.1021/acs.langmuir.4c04499
- [9] de Oliveira, R. M., Neta, M. J. M. B., de Sousa Oliveira, J., de Macedo, L. G. M., **de Menezes, R. F.**, Pirani, F., Gargano, R. (2024). Weakly bound mercury-noble gas adducts: strength, range and nature of the interaction, spectroscopic and thermodynamical properties. *Computational and Theoretical Chemistry* 1239, 114797. doi:10.1016/j.comptc.2024.114797
- [10] de Macedo, L. G. M., Negrão, C. A. B., de Oliveira, R. M., **de Menezes, R. F.**, Pirani, F., Gargano, R. (2023). The van der Waals interactions in systems involving superheavy elements: the case of oganesson (Z = 118). *Physical Chemistry Chemical Physics* 25(1), 633–645. doi:10.1039/D2CP04456K
- [11] **de Menezes, R. F.**, Pirani, F., Coletti, C., de Macedo, L. G. M., Gargano, R. (2022). Functionalized graphene-based quantum dots: promising adsorbents for CO, NO₂, SO₂, and NH₃ pollutant gases. *Materials Today Communications* 31, 103426. doi:10.1016/j.mtcomm.2022.103426
- [12] **de Menezes, R. F.**, de Macedo, L. G. M., Martins, J. B. L., Pirani, F., Gargano, R. (2021). Investigation of strength and nature of the weak intermolecular bond in NH₂ radical-noble gas atom adducts and evaluation of their basic spectroscopic features. *Chemical Physics Letters* 769, 138386. doi:10.1016/j.cplett.2021.138386

Preprints

- [13] White, K. P., Wyatt, K. W., Kerner, R. A., **Ferreira de Menezes, R.**, Chaney, T. P., Zhang, Z., Radicchi, E., Alphonso, N., Levin, A. J., Ladd, D. M., Díaz, M. M., Kitsu Iglesias, L., Dayton, D. G., Hoang, T., Weadock, N. J., Li, R., De Angelis, F., Sprenger, K. G., Berry, J. J., Toney, M. F. (2026). Solution speciation in lead iodide-based perovskite precursor inks. *ChemRxiv* preprint. doi:10.26434/chemrxiv.15006904/v1

Conference Abstracts

- [14] Porter, J. M., Landetta, J., Diehl, B., Maratos, J., **Ferreira de Menezes, R.**, Sprenger, K., Meyer, L., Brow, R., Colclasure, A. M. (2026). A quantitative infrared method for individual polysulfide speciation in lithium-sulfur electrolytes. *ECS Meeting Abstracts* MA2026-01, 267. doi:10.1149/ma2026-013267mtgabs
- [15] Allan-Cole, E., Garman, R., Roberts, L., Hartin, S., Narasimhan, S., Uselton, K., **de Menezes, R. F.**, et al. (2024). Under pressure: exploring the link between uniform stack pressure and electrode microstructure to increase performance in extreme fast charging lithium-ion batteries. *PRiME 2024, Joint International Meeting of ECS, ECSJ, and KECS* Honolulu, HI, Oct 2024.
- [16] Allan-Cole, E., Garman, R., Roberts, L., Hartin, S., Uselton, K., **de Menezes, R. F.**, et al. (2024). Effects of uniform stack pressure as a function of electrode thickness in extreme fast charging lithium-ion batteries via operando X-ray diffraction. *245th ECS Meeting* San Francisco, CA, May 2024.

Invited Talks and Presentations

Computational Materials Science and Engineering Gordon Research Conference, <i>invited talk</i>	Newry, ME, USA, Aug 2026
Batteries Gordon Research Conference, <i>invited talk</i>	Ventura, CA, USA, Feb 2026
74th Lindau Nobel Laureate Meeting, <i>young scientist participant</i>	Lindau, Germany, Jun 2025
Sandia Day, <i>poster</i>	Boulder, CO, USA, Sep 2024
Front Range Engineering-Industry Symposium, <i>poster (Best Poster Award)</i>	Boulder, CO, USA, Jul 2024
Batteries Gordon Research Conference, <i>poster</i>	Ventura, CA, USA, Feb 2024
Rocky Mountain Solid State Chemistry Workshop, <i>poster</i>	Boulder, CO, USA, Jan 2023
Federal District Scientific Initiation Congress, <i>talk (honorable mention, best presentation in natural sciences)</i>	Brasilia, Brazil, Sep 2021

Fellowships, Honors, and Grants

Lindau Nobel Laureate Meeting, <i>nominated participant (U.S. National Academy of Sciences)</i>	2025
Link Foundation Energy Fellowship, <i>honorable mention</i>	2025
Best Poster Presentation, Front Range Engineering-Industry Symposium (University of Colorado Boulder)	2024
DOE VTO award DE-FOA-0002893, Mechanistic Modeling of Li-S Batteries (<i>contributing researcher</i>)	2023 – Present
DOE VTO award DE-LC-000L096, Mechanistic Studies of the Cathode-Electrolyte Interface (<i>contributing researcher</i>)	2022 – Present
Tutorial Education Program (PET) Fellowship (Brazilian National Fund for the Development of Education, FNDE)	2018 – 2021
Silver Medal, Brazilian Physics Olympiad (Brazilian Physical Society)	2016
Bronze Medal, Regional Chemistry Olympiad (University of Brasilia)	2016
High School Merit Scholarship (Centro Educacional Leonardo da Vinci)	2013 – 2016

Outreach and Science Communication

Smithsonian National Air and Space Museum	Washington, DC, USA
Volunteer Educator	2026 – Present
– Lead hands-on activities for K-8 visitors on satellites, rocketry, museum label writing, and conservation practice.	
National Book Festival, STEM District	Washington, DC, USA
Volunteer	2026
– Supported an interactive space that pairs STEM concepts with storytelling and literary activities for young readers.	
Scientific Animations and Explainers	res07.github.io/animations
Author	2026

- Wrote and rendered a set of teaching animations on battery operation, molecular dynamics, DFT, and TDDFT for lectures and public talks.

Physics Week, University of Brasilia

Brasilia, DF, Brazil

Organizer

2018 – 2021

- Organized annual week-long programs of lectures and short courses in physics for more than 500 participants per year.

Computational Quantum Chemistry Methods Winter Course

Brasilia, DF, Brazil

Organizer

Jul 2021

- Coordinated a two-week course with the Institute of Physics at the University of Brasilia.

Teaching

University of Colorado Boulder, Department of Chemical and Biological Engineering

Boulder, CO, USA

Teaching Assistant

2023 – 2025

- Physical Chemistry, lead teaching assistant (Spring 2025): delivered lectures, wrote homework and exam problems, and led office hours and review sessions for 80 students in thermodynamics and quantum chemistry.
- Materials Characterization for Engineering (Spring 2024): X-ray diffraction, X-ray imaging, and spectroscopic methods.
- Fluid Mechanics (Spring 2023): wrote and graded homework and exams, held office hours and review sessions for 80 students.

University of Brasilia, Institute of Physics

Brasilia, DF, Brazil

Teaching Assistant and Instructor

2019 – 2021

- Teaching assistant for Classical Mechanics, Calculus, and Modern Physics Laboratory: prepared laboratory experiments, lectures, exams, and problem sets; led weekly problem-solving sections.
- Introduction to Physics (Spring 2021): designed and taught a calculus-fundamentals course for first-year students to improve retention in the physics program.

Professional Service

Ad hoc reviewer for *Royal Society of Chemistry* and *Springer journals*